

Factors in Maximal Hypertrophic Development

Several population-specific factors affect skeletal muscle mass and the hypertrophic response to resistance exercise. Of particular note are genetics, age, sex, and training experience. This chapter provides an overview of these factors and their effects on the ability to increase muscle size.

Genetics

A theoretical upper limit to muscle fiber size exists, which is ultimately determined by a person's genotype and phenotype. *Genotype* can be broadly defined as the genetic makeup of an organism; *phenotype* refers to how genotypes are expressed. In short, genetically coded information (genotype) is interpreted by the body's cellular machinery to produce the physical properties of the muscle (phenotype). With respect to hypertrophy, someone may have the genetic makeup to become an elite bodybuilder, for example, but if he or she never engages in a regimented resistance training program, that genotype will not be expressed to bring about a championship-caliber physique.

The manifestation of muscle genotype and phenotype has been extensively researched. Research in identical twins shows that up to 90% of the variance in baseline muscle mass is hereditary (33), and stark interindividual hypertrophic differences are seen in response to a resistance training program. In a study of over 500 subjects, Hubal and colleagues (36) demonstrated highly dissimilar responses in both men and women to 12 weeks of pro-

gressive resistance training of the elbow flexors. Some subjects increased biceps brachii cross-sectional area by up to 59%, while others showed little to no muscular gains. Similarly, a cluster analysis by Bamman and colleagues (7) categorized a group of young and old men and women based on their response to 16 weeks of multiset progressive lower-body resistance exercise: The top quartile increased muscle cross-sectional area by 58%, and the bottom quartile showed no mean gains; the balance of the group showed a moderate response with an increase of 28%. These findings have led to classifying subjects as *responders* and *nonresponders* to resistance exercise, thereby highlighting the role of genetics in muscle development.

Early work investigating the effects of body build on training-induced hypertrophy showed that those with a "solid" build achieved greater increases in fat-free mass compared to slender individuals following performance of a 12-week regimented resistance training program (88). However, subsequent research shows that baseline variations in muscle mass tend to be a poor predictor of hypertrophic increases induced by exercise. This indicates that a different set of genes influences variability in muscle mass attained during normal growth and development compared to that acquired from repeated bouts of mechanical overload (40). In addition, the magnitude of hypertrophic changes can vary between fiber types; greater increases in slow-twitch muscles (e.g., soleus) do not necessarily predict superior adaptations in fast-twitch muscles (e.g., plan-

KEY POINT

A variety of genetic factors influence hypertrophic potential, and this influence declines with advancing age.

taris) and vice versa (40). Moreover, the body of evidence suggests that genetics contributes less to muscular phenotype with advancing age (83).

An array of hereditary factors are believed to influence hypertrophic potential. Pioneering multidisciplinary work published in a large exercise genomics study titled “Functional Single Nucleotide Polymorphisms Associated With Human Muscle Size and Strength” (FAMuSS) identified 17 genes believed to explain some of the variances in interindividual muscular adaptations (59). One such gene, bone morphogenetic protein 2 (BMP2), is believed to be especially relevant to hypertrophic outcomes. Devaney and colleagues (20) found that polymorphisms of the BMP2 gene were responsible for differences in muscular adaptations to intense exercise. Specifically, young males with the CC genotype displayed greater gains in muscle mass following 12 weeks of progressive resistance training compared to those carrying the A allele (a form of a gene). BMP2 was estimated to explain 3.9% of the trait variation. Polymorphisms (variants) of the angiotensin-I converting enzyme (ACE) and α -actinin-3 (ACTN3) genes, among others, also have been implicated in exercise-induced muscle development (23).

The extent of hypertrophy also has been genetically linked to several growth and inflammatory factors. The ability to induce gene expression of MGF, the local form of IGF-1, appears to be particularly important in this regard. Bamman and colleagues (7) found that MGF was differentially expressed across a varied group of men and women: Extreme hypertrophic responders displayed a robust increase in MGF mRNA, whereas nonresponders experienced only a nonsignificant trend for an increase. Interestingly, genetic differences in the expression of the IGF-1Ea isoform did

not have an effect on gains in muscle mass, although other studies suggest a possible role (59). With respect to inflammatory factors, research has focused on interleukin-15 (IL-15), a myokine that has been shown to be anabolic in both in vitro and animal models. Riechman and colleagues (68) reported that a polymorphism in the IL-15 gene explained a significant proportion of the hypertrophic variation in a group of 153 young men and women following 10 weeks of heavy resistance training. These findings are supported by a recent study showing an upregulation of IL-15R α (a receptor that regulates IL-15 signaling) gene expression after resistance exercise, with a positive correlation ($r = .66$) seen between elevations and increases in myofibrillar protein synthesis (58). However, a larger trial found associations between IL-15 and baseline muscle size but no correlation in muscular adaptations to regimented resistance training (64). Findings from the latter study are consistent with recent research showing that IL-15 promotes changes more indicative of an oxidative phenotype as opposed to regulating increases in muscle mass in humans (65). Discrepancies in evidence highlight the complexities involved in determining the role of genetics in human muscular development.

There is compelling evidence that individual variances in satellite cell response play a role in a person’s hypertrophic potential. A cluster analysis of 66 untrained men and women found that extreme hypertrophic responders to resistance exercise had a greater population of satellite cells at baseline and were better able to expand the available satellite cell pool during training than modest responders and nonresponders (62). Moreover, the extreme responders were most adept at incorporating new nuclei in existing myofibers. These findings are in line with recent research showing that the acute satellite cell response to a bout of resistance training is predictive of long-term hypertrophic outcomes (8).

Emerging research indicates that micro RNAs (miRNAs) may play a significant role in the interindividual response to resistance exercise. Micro RNAs are short, noncoding RNA molecules capable of altering the translation of protein-coding genes (19). Not only do miRNAs

help to fine-tune gene expression patterns, but they also can serve as on–off switches in gene expression (95). To date, hundreds of miRNAs have been identified, and many are known to be responsive to extracellular stimuli, such as physical exercise, and thereby regulate muscle phenotype (9, 19). Davidsen and colleagues (19) found a moderate correlation between resistance training–induced muscle growth and changes in the quantity of miRNAs. Specifically, low responders presented a downregulation of miR-378, -26a, and -29a, and an upregulation of miR-451; these changes were linked to a suppression of mTOR signaling. Additional miRNAs that have been linked to hypertrophic adaptations include miR-1, miR-29c, miR-128a, and miR-133a/b, among others; the influence of genetics across the full spectrum of miRNAs remains to be fully explored. The collective findings suggest a hereditary link between certain miRNAs and human skeletal muscle hypertrophy.

Muscle morphology is another potential candidate for genetic differences in the hypertrophic response to resistance training. Cadaver studies show significant interindividual differences in fiber number between individuals (2). By the age of 24 weeks, fiber numbers remain constant; further increases in growth are attributed almost exclusively to hypertrophy as opposed to hyperplasia (83). Logically, a greater number of fibers would be advantageous to increasing muscle size. Research lends support to this hypothesis, as a moderate correlation has been noted between fiber number and whole-muscle cross-sectional area. Moreover, a group of male bodybuilders and age-matched controls showed that those with the largest biceps brachii had a larger number of fibers in this muscle (48).

Differences in muscle fiber type may also play a role in the phenotypic response to resistance training. Approximately 45% of the variance in fiber type is thought to be associated with genetic factors (78). Substantial heterogeneity exists in fiber type percentages between individuals; for example, approximately 25% have either less than 35% or more than 65% Type I fibers in the vastus lateralis muscle, with a reported range of 5% to 90% (78).

Moreover, dominance of a given fiber type in a given muscle is not necessarily indicative of whole-body fiber type proportions; those with a high percentage of Type I fibers in one muscle could have a high percentage of Type II fibers in another muscle. The prospect that variances in fiber type percentage could be responsible for differential hypertrophic adaptations seems to have a logical basis. Fast-twitch fibers grow about 50% more than their slow-twitch counterparts following resistance training, although a high degree of interindividual variability is seen with respect to the extent of hypertrophic adaptations (42). Anecdotally, athletes with higher percentages of Type II fibers are more muscular in appearance than those dominant in Type I fibers. Interestingly, a cluster analysis revealed that the degree of hypertrophy in response to regimented resistance training did not differ on the basis of pretraining percentages of Type I and Type II myofibers (7). However, Haun and colleagues (35) recently provided contradictory evidence on the topic, showing that pretraining Type II fiber percentage was a strong predictor of hypertrophic gains in a cohort of trained men performing a 6-week resistance training program. Additionally, results showed that the highest hypertrophic responders tended to possess lower pretraining Type II cross-sectional areas, potentially indicating a higher ceiling for growth.

Although it is tempting to look at genes in isolation, it is likely that interactions of multiple genetic loci (the specific location of a gene, DNA sequence, or position on a chromosome) ultimately determine a person's genetic capacity (59). The hypertrophic impact of a single genetic influence tends to be fairly modest,

KEY POINT

Although the terms *responders* and *nonresponders* have been proposed in the literature, even nonresponders can significantly increase muscle mass over baseline levels. However, they may require longer periods of consistent training and alternative training strategies to gain additional hypertrophy.

but the combination of variances can have a profound effect on phenotype. Moreover, the term *nonresponder* is somewhat of a misnomer. Although approximately 25% of subjects show little to no growth following a research-based resistance training protocol (7), this does not necessarily imply that these people are inca-

pable of increasing muscle mass. The duration of most resistance training studies is relatively short—usually a few months. Anecdotally, the overwhelming majority of those who train consistently for long periods ultimately gain significant muscle mass, albeit less than responders (16). In addition, just because a person fails to

RESEARCH FINDINGS

DO MUSCLES HAVE AN EPIGENETIC MEMORY?

Much like the brain, skeletal muscles are said to have a “memory” that allows them to recall previous mechanical events. Skeletal muscle memory refers to both cellular and tissue retention of prior stimuli (e.g., stress from exercise) that leads to a modified response if the stimulus is reencountered (77). While traditionally the concept of muscle memory applied to relearning a motor task, recent evidence indicates that it also may have relevance to hypertrophy.

In chapter 1, we discussed that satellite cells provide memory for muscles, and hypertrophy lost through detraining is regained when training resumes because of the retention of myonuclei that facilitate a greater transcriptional potential of the fibers. It has been hypothesized that muscles also possess an epigenetic (translated as “above genetics”) memory that further enhances hypertrophic adaptations after reintroduction to anabolic stimuli. Epigenetics can be operationally defined as changes in the activity and expression of genes brought about by structural cellular modifications without altering the genetic code (77). These modifications are primarily specific to DNA and histones (e.g., methylation and acetylation), but they also can apply to posttranscriptional alterations of RNA. An attenuation in DNA methylation of genes mediates enhancements in gene expression because removing methylation affords greater access to the machinery that facilitates gene transcription (75).

An emerging body of research supports the concept of epigenetic muscle memory. Acute exercise demethylates various promoters of given genes, resulting in expression of the associated genes. Demethylation is specific to the intensity of aerobic exercise, with higher intensities targeting genes such as PPAR- γ , PGC-1 α , PDK4, and MEF2A; these effects are seen immediately after exercise and, in some cases (such as with PPAR- δ), 3 hours post-workout (55). What is most interesting is that muscles apparently are able to retain this molecular information and use it later when faced with the same exercise stressor to facilitate appropriate adaptations.

Seminal work from the lab of Adam Sharples provides evidence that epigenetic memory extends to hypertrophic adaptations obtained from resistance training (75). Employing a within-subject design, untrained men performed 7 weeks of regimented total-body progressive resistance exercise carried out 3 days a week. This was followed by a 7-week detraining period in which no exercise was performed. Subjects then reengaged in the same exercise for an additional 7 weeks. Results indicated that various hypertrophy-related genes in the trained muscles remained in a hypomethylated state during detraining, and their expression was switched on to an even greater extent upon retraining. Several genes in particular showed significantly enhanced expression upon reloading (RPL35a, UBR5, SETD3, and PLA2G16), and their expression was highly correlated with the change in lean mass.

(continued)

Do Muscles Have an Epigenetic Memory? *(continued)*

The UBR5 gene appears especially relevant to exercise-induced hypertrophy because it has been found to be involved at the DNA-methylation, gene, and protein level in recovery and growth across human, mouse, and rat studies. Human genetic association research (examining over 700,000 single nucleotide polymorphisms across the genome) demonstrates that certain polymorphisms of the UBR5 gene are strongly associated with a larger cross-sectional area of fast-twitch muscle fibers, and occur more frequently in strength and power athletes compared with endurance athletes and untrained individuals (76). Although speculative, the findings suggest that individuals possessing these UBR5 polymorphisms may have both a genetic and an epigenetic propensity for muscle memory.

Taken with evidence that satellite cells also possess a “memory,” the findings highlight the unique capabilities of muscle to respond and adapt to stimuli. Importantly, these adaptive capabilities are constantly evolving, suggesting that a person’s hypertrophic responsiveness is, at least in part, predicated on previous training experience. It also provides a basis for speculation that brief periods of reduced loading do not negatively affect growth, and in fact may present a strategy to resensitize the anabolic capacity of muscle and thus spur future hypertrophic gains.

respond to one training protocol does not necessarily mean that he or she will not respond to an alternative protocol. For example, it has been postulated that a fiber type–specific approach to training may enhance the genetic capacity to hypertrophy. Specifically, people dominant in Type I fibers may obtain superior results from training with lighter loads, whereas those dominant in Type II fibers would be best served by employing heavy loads (26). This hypothesis warrants further investigation. Moreover, some people respond better to lower training volumes and frequencies (62), suggesting that genetic limitations can be surmounted, at least in part, by manipulating both of these variables over time.

It should be noted that the genetic predisposition to hypertrophic gains can be specific to a given muscle. A common complaint from those who resistance train is the difficulty in bringing up a lagging muscle group. Indeed, observations from studies carried out in my lab routinely see one subject showing significant increases in quadriceps growth with little to no growth in the elbow flexors and another subject displaying the opposite growth pattern. Again,

this does not necessarily reflect an inability to increase muscle size in the lagging muscle, but rather the need to employ alternative training strategies to spur additional hypertrophy.

Age

The aging process is associated with alterations in both the quantity and quality of muscle. Human muscle mass reaches peak levels between the ages of 20 and 40 (14). Thereafter, the body loses approximately 0.5% of its muscle mass per year during the fourth decade of life, increasing to 1% to 2% annually after the age of 50 and then accelerating to 3% annually after the age of 60 (figure 7.1) (91, 97). This age-related loss of muscle tissue has been termed *sarcopenia*. Sedentary people show larger rates of decline than those who are active, although leisure time physical activity has only minor effects on tempering muscle loss (91). Sarcopenic changes have been attributed to reduced rates of basal, postabsorptive myofibrillar muscle protein synthesis or elevated proteolysis, or both, but more recent findings suggest that basal skeletal muscle net protein

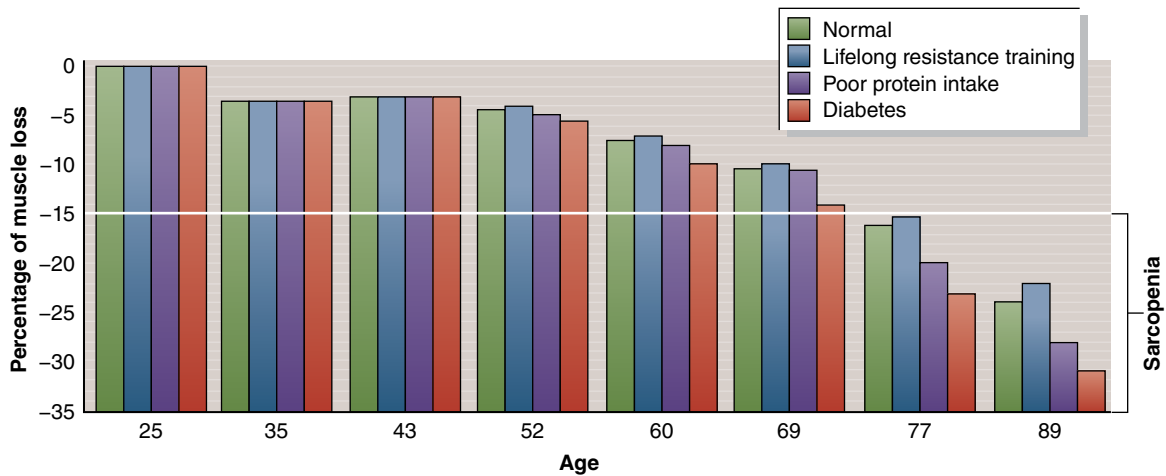


FIGURE 7.1 Rate of muscle mass loss with age.

Data from Buford et al. (12).

balance is not compromised with aging in healthy people (11). Alternatively, it has been postulated that chronic systemic inflammation may compromise muscle protein metabolism in the frail elderly (11). Various disease states and lifestyle factors are known to exacerbate the rate of muscle wasting with age.

Sarcopenia is characterized not only by fiber atrophy, but also by widened sarcoplasmic spaces and Z-band and myofibrillar disruption (79). These negative effects are seen in both Type I and Type II fibers, but they are most pronounced in the fast-twitch variety. There is evidence that Type II fibers actually undergo *apoptosis* (programmed cell death as part of normal growth, development, or aging). The number of these fibers decreases from 60% in sedentary young men to less than 30% in people over the age of 80 (24). Autopsy results show that the quadriceps muscles in the elderly are 18% smaller than those in younger adults, and the total fiber number is 25% lower; a reduction of approximately 110,000 fibers is attributed to the aging process (44). Other research indicates a significant decline in the number of myofibers regardless of fiber type between the sixth and eighth decades of life (45). In addition, an alteration in the chemical and physical properties of skeletal muscle proteins occurs, which includes reduced contractile, mitochondrial, and enzyme protein synthetic rates; altered expression and post-

translational modifications to muscle proteins; reduced maximal voluntary muscle strength; and reduced muscle strength per unit of muscle mass and muscle power (96). These changes are apparently mediated, at least in part, by a chronic decrease in circulating levels of testosterone, GH, and IGF-1 (13). Sarcopenic changes in myofibers are accompanied by deleterious structural alterations to the extracellular matrix, which further impairs muscle tissue remodeling (29).

Satellite cell content is also altered as one ages, particularly in Type II muscle fibers. The number of satellite cells per Type II fiber has been shown to be markedly lower in the elderly than in the young, as are the number of satellite cells relative to total nuclei (89). A number of other studies support these findings (38, 67), although some have failed to show significant differences in satellite cell populations (70). Null findings have been attributed to a lack of muscle fiber type-specific data (89). In addition, satellite cells from older muscles fail to activate and proliferate when subjected to muscle injury, demonstrating an impaired self-renewal from aging (22). Taken as a whole, the body of evidence strongly indicates that the age-related atrophy of Type II fibers is associated with a fiber type-specific decline both in satellite cell content and their ability to respond to stimuli, which would likely accelerate the extent of sarcopenic changes.

Regular resistance training can attenuate muscle loss in the elderly and, depending on genetic, environmental, and training-related factors, even produce increases in lean mass above that in sedentary younger people. However, the hypertrophic potential is blunted with advancing age. This anabolic insensitivity is reflected in the acute response to resistance training. Kumar and colleagues (43) found that phosphorylation of p70^{S6K} and eIF4EB1 at 60% to 90% of 1RM was diminished in older men following multiple sets of unilateral knee extension and flexion exercises at 60% to 90% of 1RM. Moreover, p70^{S6K} phosphorylation was uncoupled with the rate of muscle protein synthesis at 1 to 2 hours post-exercise in elderly subjects, but not in the young. Other studies show similar findings (27, 43, 92). The totality of evidence indicates an age-induced anabolic resistance of intracellular signaling and muscle protein synthesis to resistance exercise.

Most longitudinal research studies support the notion of a diminished hypertrophic response to resistance exercise in the elderly (42, 50, 52, 93), although some studies show no age-related differences in muscle protein accretion (32, 71). It appears that the time course of muscle growth is altered in aging, with evidence of a delayed hypertrophic response in the early stages of resistance training (46, 85). Moreover, a substantially greater percentage of the elderly are deemed nonresponders to resistance exercise compared to young subjects (7). The underlying reasons for the age-related impairment of muscular adaptations are not entirely clear, but alterations in chronic anabolic hormonal profiles appear to play a causative role (54). Other potential mediating factors include a combination of anabolic resistance, chronic low-grade systemic inflammation, compromised satellite cell function, reduced angiogenesis, and blunted ribosome biogenesis. That said, older adults can and do see robust muscle growth after performing regimented progressive resistance training protocols. Hypertrophic gains in excess of 20% are routinely seen in this population, and increases are noted in both Type I and Type II muscle fibers (7). Even the very elderly (≥ 75 years of age) respond favorably to resistance training; increases in cross-sectional

KEY POINT

After age 40, the body loses progressively more muscle mass per year. Regular resistance training can reduce this loss. Although the elderly do show a diminished hypertrophic response, they can gain muscle mass; however, a greater weekly training dose appears necessary to maintain these gains.

area of 1.5% to 15.6% have been reported in the literature (84). Meta-analytic data indicate that moderately higher training volumes become increasingly beneficial to maximize muscle mass as we age (61).

Research indicates that the aging process results in an impaired recovery following exercise, and several studies show that it takes longer for older individuals to restore performance to baseline levels compared to younger trainees for a similar exercise stimulus (25). It is speculated that these impairments may be related to a greater exercise-induced muscle damage or heightened fatigue response, or both. Regardless of the mechanisms, evidence suggests that the elderly may benefit from fewer weekly training sessions to allow for regeneration of neuromuscular capacity; alternative strategies also may be considered to facilitate restoration following exercise (e.g., nutritional supplementation, massage). Although large interindividual differences in recuperative abilities between older trainees exist, overall it appears clear from the literature that more attention must be directed to managing recovery in this population.

Older individuals also may not be able to tolerate as much volume as their younger counterparts. Support for this hypothesis can be gleaned from a study by Bamman and colleagues (7), who investigated the response to regimented resistance training in a cohort of 66 men and women, with an approximately equal distribution of younger and older subjects. Participants performed a 16-week training program consisting of the squat, leg press, and leg extension. All subjects performed 3 sets of each exercise, 3 days a week, for a total 27 sets per week of lower-body

exercise. Cluster analysis showed that the vast majority of those considered nonresponders were older individuals; conversely, few elderly subjects were categorized as extreme responders; these observations were reversed for the younger subjects. Although volume was not isolated as an independent variable, the findings suggest that the protocol may have been too demanding for the older subjects. Although specific recommendations cannot be determined from the literature, training volume should be scrutinized as people age, with possible reductions required both on a per-session and per-week basis.

Alternatively, research by Bickel and colleagues (10) indicates that elderly people need a greater weekly minimum training dose to maintain muscle once they have achieved a given level of hypertrophy from resistance training. Seventy young (20 to 35 years of age) and old (60 to 75 years of age) participants performed a 3-day-per-week resistance training program for 16 weeks. Following training, the subjects were randomly assigned to a detraining protocol involving no exercise, a maintenance protocol that was 1/3 that of the original program, or a maintenance protocol that was 1/9 that of the original. As expected, progressive resistance training resulted in significant hypertrophic increases in both the young and the old. However, although the two maintenance protocols were sufficient for preserving hypertrophy in the young, the elderly in both maintenance groups showed significant reductions in muscle size.

Sex

Substantial sex-based differences exist in the maintenance and hypertrophy of skeletal muscle tissue. On average, women have less muscle mass than men from both an absolute and relative standpoint. In support of this fact, men maintain approximately 10 kg (22 lb) more lean mass compared to women at any given body weight (69). These discrepancies become evident during puberty and persist through old age.

It is believed that sexual dimorphism is highly influenced by hormonal variances between the sexes. Testosterone levels in men are approximately 10 times higher than those

in women. As discussed in chapter 1, testosterone is a highly anabolic hormone that exerts its actions by increasing myofibrillar protein synthesis and decreasing muscle protein breakdown (87, 98). Theoretically, low circulating testosterone levels in women would reduce the potential to substantially increase muscle mass. However, attenuations in anabolism from a lack of testosterone appear to be at least partially offset by higher estrogen levels. The anabolic effects of estrogen are attributed to reductions in muscle protein breakdown; a hypothesis supported by research showing that hormone replacement therapy counteracts the upregulation of the ubiquitin–proteasome system in menopausal women (66). There also is evidence that estrogen positively modulates myogenic gene expression following resistance training, indicating a potential role in enhancing sensitivity to anabolic stimuli (21).

On a relative basis, men and women experience similar increases in muscle hypertrophy following regimented resistance training (1, 36, 42). However, these results must be understood in the context that women start off with less muscle mass at baseline, thus biasing increases in their favor. From an absolute standpoint, hypertrophic gains are significantly greater in men than in women. Ivey and colleagues (37) found that men increased muscle volume approximately twice as much as women following 9 weeks of unilateral knee extension exercises. In a study of elite bodybuilders, biceps brachii cross-sectional area was two times larger in male than in female competitors (4). These sex-based differences were primarily attributed to greater absolute mean Type II fiber areas in male bodybuilders. Males also had a greater total number of muscle fibers, a finding that

KEY POINT

Although men and women experience similar relative increases in muscle hypertrophy following regimented resistance training, men achieve significantly greater absolute gains, which is seemingly attributed, at least in part, to their higher testosterone levels.

has been reported in other studies as well (72). So although women can build appreciable muscle from regimented resistance exercise, their hypertrophic potential is somewhat less on average than that of men.

Aging appears to have a particularly detrimental effect on muscle mass in women (figure 7.2). Despite higher resting protein synthetic rates in the postmenopausal period, elderly women experience an accelerated loss of muscle resulting from increased rates of proteolysis, a phenomenon partly attributed to decreased estrogen production (34). Moreover, the anabolic response to protein feeding is blunted to a greater degree in older women (80). In addition, the hypertrophic response to resistance training is impaired in elderly women (6, 42), as are post-exercise elevations in muscle protein synthesis (81). Indeed, females display higher frailty index

scores than males across every age group (30), as well as possessing a lower Type II fiber size, satellite cell content, and myonuclear domain (39). Taken together, these findings indicate that postmenopausal reductions in estrogen in women have a more detrimental impact on muscle mass than decreased testosterone levels associated with aging in men.

Despite these obstacles, elderly women can significantly increase fundamental muscle mass with regimented resistance exercise (15, 57, 90). Training-induced increases in hypertrophy have been correlated with reductions in primary inflammatory markers such as C-reactive protein (CRP) and tumor necrosis factor- α (TNF- α) (57). Whether a cause-effect relationship exists is not clear, but these correlations raise the possibility that chronic inflammation is particularly detrimental to older women in their ability to build muscle. Older women

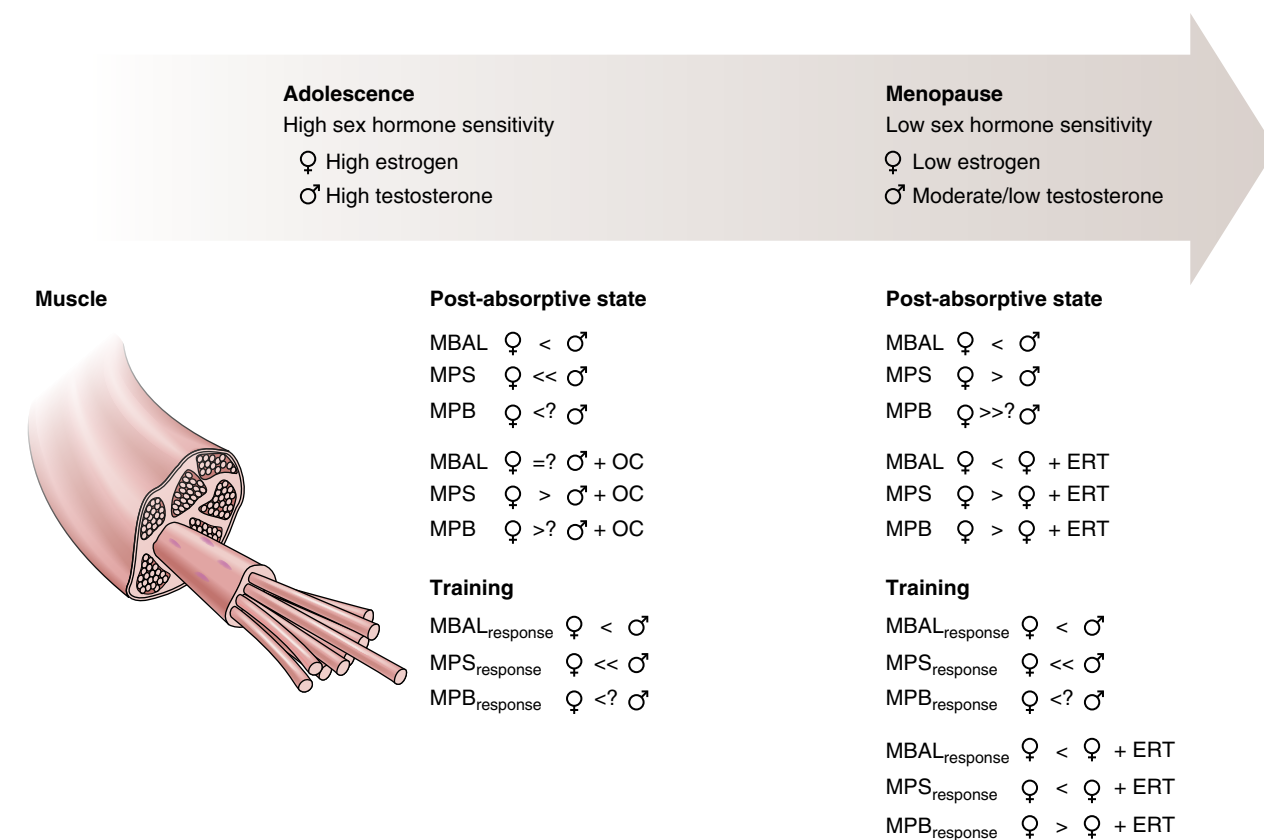


FIGURE 7.2 Effect of menopause on hypertrophic development. MBAL = muscle protein balance; MPS = muscle protein synthesis; MPB = muscle protein breakdown.

Reprinted by permission from M. Hansen and M. Kjaer, "Influence of Sex and Estrogen on Musculotendinous Protein Turnover at Rest and After Exercise," *Exercise and Sport Sciences Reviews* 42, no. 4 (2014): 183-192.

also display a blunted hyperemic response to exercise compared to older men, which may impair amino acid delivery to the working muscles and thus attenuate the training-induced anabolic response (82). This raises the possibility that performing supplementary aerobic training may be an effective countermeasure to the issue because it can help to enhance angiogenesis and thereby potentially facilitate nutrient transport.

In regard to exercise performance, evidence suggests that women display faster recovery following a resistance training set than their male counterparts (31). It is not clear whether this is because women tend to train with lighter loads than men, or whether other sex-related factors come into play. Regardless, women may thus be able to employ somewhat shorter rest intervals without compromising muscular development. At the very least, this allows for a greater training efficiency, reducing the time required to optimize results.

Training Status

The vast majority of resistance training studies are carried out in untrained individuals. This is generally a function of convenience because the pool of untrained subjects is larger than the pool of resistance-trained subjects. However, the hypertrophic response of trained subjects is substantially different than that of their untrained counterparts (60), thereby limiting the generalizability of such studies outside of the initial stages of training.

Differences in the hypertrophic potential between trained and untrained people can be attributed to the *ceiling effect*, or *window of adaptation* (figure 7.3). During the initial stages of training, the neuromuscular system is deconditioned and responds to virtually any stimulus because the ceiling for growth is high. Even steady-state cardiorespiratory exercise has been shown to produce hypertrophic increases in previously sedentary individuals (41). As people become resistance trained and move closer to their genetic ceiling, however, it becomes progressively more difficult to increase muscular size (i.e., the window of adaptation becomes smaller). Theoretically, an

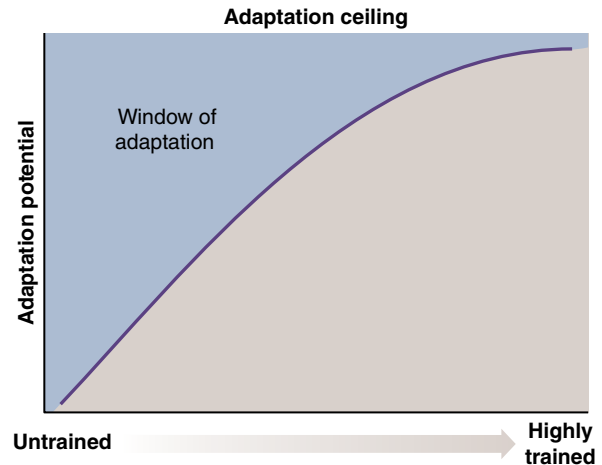


FIGURE 7.3 The ceiling effect, or window of adaptation.

excess of muscle mass would be energetically and kinetically inefficient, and thus the human body limits the amount of lean tissue that can be gained. In support of this hypothesis, research shows that the extent of hypertrophic gains is relatively small (approximately 3% to 7%) in highly competitive bodybuilders over 5 months of resistance training, suggesting these people are at the upper limits of their genetic ceilings (5).

Alterations in anabolic intracellular signaling have been demonstrated between trained and untrained subjects in both animal and human models. Ogasawara and colleagues (56) exposed male rats to maximal isometric contractions via percutaneous electrical stimulation of the gastrocnemius muscle every other day for either 1 bout, 12 bouts, or 18 bouts. Those in a detraining group performed 12 bouts, detrained for 12 days, and then were subjected to an additional exercise session prior to being sacrificed. Phosphorylation of p70^{S6K}, ribosomal protein S6, and p90^{RSK} were elevated in the group that performed 1 bout, but repeated exercise bouts suppressed phosphorylation levels. This suggests that anabolic signaling becomes desensitized to resistance training when it is performed consistently over time. In a human study, Coffey and colleagues (17) investigated the effects of multiple sets of maximal isokinetic knee extensions in well-trained cyclists versus competitive pow-

erlifters. Post-exercise biopsy results showed that AMPK was significantly elevated in the aerobic endurance-trained subjects, but not the strength-trained subjects. Moreover, p70^{S6K} and S6 ribosomal protein phosphorylation was markedly elevated in the aerobic endurance-trained subjects, but not strength-trained subjects. Similarly, Wilkinson and colleagues (94) found that the duration of elevations in Akt and p70^{S6K} phosphorylation was attenuated, and the levels of S6 phosphorylation remained similar to resting levels after 10 weeks of resistance training. Other research has reported that well-trained weightlifters and powerlifters demonstrate suppressed phosphorylation of ERK 1/2, an important anabolic signaling pathway, compared to sedentary controls after performance of a chronic resistance training program (28). These results are consistent with evidence showing that genes involved in cellular hypertrophy are suppressed following a regimented resistance training protocol (53). That said, other research contradicts these findings, leading to speculation that nutritional strategies, particularly those including higher intakes of protein, may enhance training-induced hypertrophic increases in well-trained individuals (47).

Similar to the findings of acute signaling studies, there is evidence that the muscle protein synthetic response to resistance exercise is blunted in well-trained lifters. Whereas muscle protein synthesis remains elevated in the untrained state for 48 to 72 hours (51, 63), research indicates that the time course is truncated in trained subjects (their levels return to baseline within approximately 36 hours) (49, 86). It should be noted, however, that substantial individual variation exists in this response, and elevations in muscle protein synthesis in some trained subjects can persist up to 48 hours and perhaps longer post-exercise (49). The attenuated muscle protein synthesis duration following regimented training may be related at least in part to the protective response of the repeated bout effect. Given that well-trained individuals have conditioned their muscles to the stress of resistance exercise, the associated tissue breakdown is reduced and thus there is less need for remodeling (18).

KEY POINT

As people become resistance trained and move closer to their genetic ceiling, it becomes progressively more difficult to increase muscular size. Meaningful hypertrophic responses can be gained by precise manipulation of program variables, including strategic brief periods of deloading to restore the anabolic responsiveness of trained muscle.

Longitudinal changes in the anabolic response become increasingly evident over the first few months of initiating regimented resistance training. For example, Wilkinson and colleagues (94) showed that the muscle protein synthetic response was modified over the course of a 10-week resistance training program in which myofibrillar proteins continued to be stimulated but activation of mitochondrial proteins was suppressed. These findings indicate the body rapidly shifts toward coordinating intracellular responses to promote specific exercise-induced adaptations (i.e., the SAID, or specific adaptations to imposed demands, principle). However, a lack of novelty in exercise program design inevitably slows progress as the impetus for adaptation is reduced. Hence, to sustain hypertrophic gains over time necessitates progressively challenging the neuromuscular system in a manner sufficient to stimulate fibers in a novel fashion.

It should be noted that the ceiling effect is an abstract concept. Although a theoretical hypertrophic ceiling does exist, people never actually realize their full genetic potential. The ability to further increase muscle mass is always present. Indeed, muscular gains can be made even at very advanced levels, albeit at a much slower pace than during the initial stages of training. Numerous research studies show that those with considerable training experience do build appreciable muscle when a novel stimulus is applied (3, 73, 74). The results of Alway and colleagues (5) showing modest muscle growth in competitive bodybuilders indicate that the precise manipulation of program vari-

ables becomes increasingly important to elicit a meaningful hypertrophic response as people approach their genetic ceiling for hypertrophy. Moreover, there is evidence that integrating brief periods of detraining can restore the anabolic responsiveness of trained muscle (56). It

is therefore possible that bodybuilders in the Alway and colleagues (5) study might have improved their hypertrophic response by periodizing volume and intensity over the course of the training cycle to include deload periods that facilitate remodeling and rejuvenation.

TAKE-HOME POINTS

- There is a large genetic component in the individual hypertrophic response. A wide array of genes have been identified as playing a role in the ability to gain muscle. It is likely that interactions of multiple genetic loci ultimately determine a person's genetic potential to gain muscle. Hereditary differences in muscle morphology also are believed to govern the extent of a person's muscle-building capacity. Although the terms *responders* and *nonresponders* have been discussed in the literature, these classifications are overly simplistic; virtually everyone can increase muscle mass above baseline levels with consistent resistance training over time, but the ultimate extent of hypertrophy will vary greatly between individuals.
- Biological aging has a marked effect on muscle mass. Peak mass is achieved between the third and fifth decades of life, after which a gradual, progressive loss of muscle ensues (i.e., sarcopenia). An age-related reduction in anabolic hormones and satellite cell function are believed to be largely responsible for sarcopenic changes. Chronic low-grade inflammation also appears to play a role in the process. Regular resistance exercise can help abate age-related muscle loss and even produce hypertrophic increases above that in sedentary younger people. However, hypertrophic potential diminishes with advancing age, and evidence indicates that elderly people need a greater weekly minimum training dose to maintain muscle once they have achieved a given level of hypertrophy.
- The ability to build muscle differs between the sexes. Although women attain approximately equal relative muscle growth compared to men following regimented resistance training, men gain significantly more muscle on an absolute basis. These differences are seemingly attributed, at least in part, to variances in circulating testosterone. Women tend to experience a greater age-related muscle loss than men, conceivably mediated by postmenopausal reductions in estrogen levels.
- Hypertrophic capacity progressively diminishes as people become more trained. This is attributed to a ceiling effect in which alterations in anabolic intracellular signaling impair the ability to accrete muscle proteins with consistent participation in a resistance training program. However, although a theoretical ceiling does exist, people never actually realize their full genetic potential; the ability to further increase muscle mass is always present.